Quiz 3

Graded

Student

Colin Skinner

Total Points

43.5 / 50 pts

Question 1

Honor Code Acknowledgment

0 / 0 pts

✔ **+ 0 pts** Incorrect

+ 0 pts Correct

Question 2

(no title)

5 / 5 pts

✔ **+ 5 pts** Correct

+ 4 pts Minor intermediate calculation error OR omitted entry in belief vector.

+ 3 pts Did not normalize belief vector OR multiple minor errors.

+ 1 pt Major calculation errors.

+ 0 pts Incorrect OR No answer

Question 3

(no title)

5 / 5 pts

✔ **+ 5 pts** Correct

+ 4 pts Reported belief vector instead of probability.

+ 4 pts Minor calculation error.

+ 3 pts Reported belief vector instead of probability AND minor calculation error(s).

+ 2 pts Major/persistent calculation error(s) but correct approach.

+ 1 pt Major issues in approach.

+ 0 pts Blank/Incorrect

Question 4

(no title)

5 / 5 pts

✔ **+ 5 pts** Correct

+ 4.5 pts Minor miscalculation in expected reward

+ 4 pts Failed to report or incorrectly reported which location should be sampled.

+ 2.5 pts Used incorrect values for expected reward(s).

+ 0 pts Incorrect

Question 5

(no title) **5** / 5 pts

✔ **– 0 pts** Correct

– 1 pt **Arithmetic Error**: Minor numerical error in calculations.

– 1 pt **Incorrect Normalization**: Performed the normalization step but computed the normalized values incorrectly.

– 1 pt **Missing Final Answer**: Did not compute the final probability from the belief.

– 2 pts **Incorrect Observation Model**: Used the wrong likelihood values.

– 2 pts **Missing Normalization Step**: Used unnormalized likelihoods or belief values and never computed posterior probabilities.

– 2 pts **Wrong Prior**: Did not start from the correct uniform prior.

– 5 pts Incorrect

Question 6

(no title) **1.5** / 5 pts

Computing ratio

+ 4 pts Correct.

+ 3.5 pts Correct approach, minor mistake.

+ 3 pts Correct approach, incorrect number of actions or observations.

+ 2.5 pts Correct approach, major mistake or multiple minor mistakes.

+ 4 pts Alternate solution (based on 2024 Q9): Directly applying the formula without considering s_{EM}.

✔ **+ 1.5 pts** Alternate solution (based on 2024 Q9): Directly applying the formula with an incorrect number of actions or observations.

+ 0 pts Incorrect answer.

+ 0 pts No answer.

Drawing one possible 2-step conditional plan

+ 1 pt Correct drawing.

+ 0.5 pts Partially correct drawing.

✔ **+ 0 pts** Incorrect drawing.

+ 0 pts No answer

Question 7

(no title) **2** / 5 pts

- **– 0 pts** Correct Answer
- **– 0 pts** Alternate Solution
- **– 1.5 pts** Few Incorrect Intermediate Steps
- ✔ **– 3 pts** Mostly Incorrect/Incomplete Intermediate Steps
- **– 5 pts** Completely incorrect approach/no work submitted
- **– 0 pts** Flag
- **+ 1 pt** Correct utilities.
- **+ 0.5 pts** Correct reward
- **+ 0.5 pts** Correct belief
- **– 0.5 pts** Forgot to multiply by $\gamma = 0.9$
- **– 0.5 pts** Forgot to add the reward

Question 8

(no title) **5** / 5 pts

- ✔ **+ 5 pts** Correct
- **+ 4.5 pts** Very minor miscalculation for the alpha vector
- **+ 4.5 pts** Correct alpha vector values, selected the correct policy, but did not provide a justification
- **+ 4 pts** Correct alpha vector values, but chose the incorrect policy
- **+ 3.5 pts** Correct alpha vector values, but did not attempt to give an interpretation of which policy to choose
- **+ 3 pts** Miscalculated the alpha vectors
- **+ 0 pts** Fully incorrect or missing

Question 9

(no title) **5** / 5 pts

- ✔ **+ 5 pts** Correct
- **+ 2.5 pts** Mentions only that QMDP assumes full observability
- **+ 2.5 pts** Mentions only that FIB takes the observation model into consideration (mentions partial observability, belief state or uncertainty)
- **+ 0 pts** Incorrect reasoning for both QMDP and FIB

Question 10

(no title) **5** / 5 pts

- ✔ **+ 5 pts** Correct
- **+ 4 pts** Incorrect or no mention of either QMDP or PBVI (but correct reasoning about the other)
- **+ 2.5 pts** Reasoned through the problem but incorrect answer
- **+ 0 pts** Fully incorrect or missing

Question 11

(no title)

5 / 5 pts

✔ **+ 5 pts** Correct

+ 4.5 pts Rounding error: less than two decimal places

+ 4 pts Minor arithmetic or calculation error

+ 4 pts Correctly reports action scores, but not all of them

+ 4 pts Minor error with $N(h)$ and $N(h, a)$ values

+ 3 pts Wrong UCB1 formula (e.g. 2 in square root numerator, divide $Q(h, a)$ by $N(h, a)$)

+ 3 pts Conceptually incorrect interpretation of $N(h)$ and $N(h, a)$

+ 3 pts Only reports best score, no other work shown

+ 2 pts Incorrect numbers with minimal work shown

+ 0 pts No response

Q1 Honor Code Acknowledgment

0 Points

You may consult any material (e.g., books, calculators, computer programs, and online resources), but you may not consult other people inside or outside of the class or large language models like *ChatGPT*.

The quiz is designed to be completed in 150 minutes, but we will grant you 180 minutes total to complete and submit your quiz (including uploading any images, handling any logistical issues, etc.) **The timing on Gradescope is a hard cutoff.**

Ed will not allow any public posts during that time. Correct answers will receive full credit, and partial credit may be awarded based on work shown. Hence, we encourage you to show your work. You can upload **either** a file **or** give a response in the free response field (LaTeX is possible using *two* $ signs before and after your math expression – so a total of four $ signs).

Out of fairness to all students, only material submitted during the allowed time will be graded.

As a reminder, all questions are equally weighted. Some questions will take longer than others. Feel free to answer the questions in any order you like.

Write your name in the box below.

Colin Skinner

You may download the image below for reference. It contains details that pertain to the rest of the quiz. Alternately, you can find a PDF version here: https://drive.google.com/file/d/1Jkahzik7dFIQo3OFNTCcVdEx1LKjRnQ_/view?usp=drive_link

Life on POMDPlanet?

A spacecraft has landed on POMDPlanet to search for signs of life (biosignatures). While it can't measure sample quality directly, the spacecraft receives noisy spectrometer scans to estimate the potential that a location contains a biosignature. It can sample two candidate locations, L_A and L_B. We model the mission as follows:

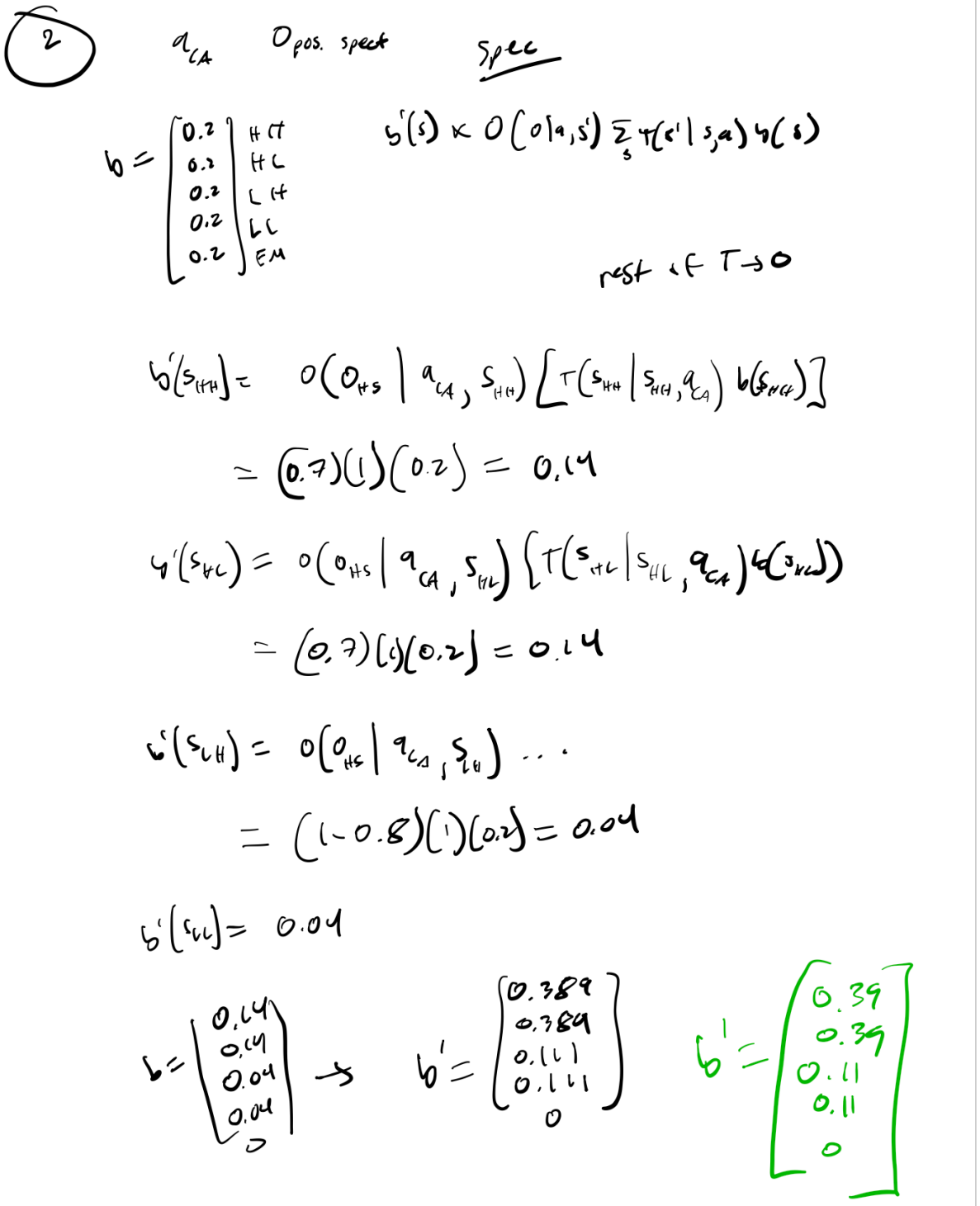
- States: Each location has either high or low quality samples. There is also a single end-of-mission state.
  - s_{HH}: L_A has high-quality samples, L_B has high-quality samples.
  - s_{HL}: L_A has high-quality samples, L_B has low-quality samples.
  - s_{LH}: L_A has low-quality samples, L_B has high-quality samples.
  - s_{LL}: L_A has low-quality samples, L_B has low-quality samples.
  - s_{EM}: The mission has ended.
- Actions: We can either scan a location, sample at a location, or end the mission.
  - a_{CA}: Scan L_A
  - a_{CB}: Scan L_B
  - a_{MA}: Sample L_A
  - a_{MB}: Sample L_B
  - a_{EM}: End the mission
- Observations:
  - If we scan (choose a_{CA} or a_{CB}), we observe either a positive or negative reading from the spectrometer.
  - If we sample (choose a_{MA} or a_{MB}), we observe either a biosignature detection or no detection.
  - If we have ended the mission (a_{EM}), we receive no observation.
  - The probability of a positive reading for a site with high-quality samples is 0.7.
  - The probability of a negative reading for a site with low-quality samples is 0.8.
  - The probability of no biosignature detection in a high-quality sample is 0.05.
  - The probability of a biosignature detection in a low-quality sample is 0.05.
- Transitions: Unless we end the mission (a_{EM}), assume states do not change over time. The action a_{EM} deterministically transitions the system to s_{EM}. Taking any action from s_{EM} keeps the system in s_{EM}. Equivalently:
  - For any action $a \neq a_{EM}$, $T(s' \mid s, a) = 1[s' = s]$.
  - $T(s_{EM} \mid s, a_{EM}) = 1$ for all s.
  - $T(s_{EM} \mid s_{EM}, a) = 1$ for all a.
- Rewards:
  - If we sample a site and it has high-quality samples, we get a reward of $+50$.
  - If we sample a site and it has low-quality samples, we get a reward of -10.
  - If we scan a site, we always get a reward of -1 (to account for power draw).
  - If we end the mission, we get a reward of 0.
- Discount Factor (γ): $\gamma = 0.9$

Q2

5 Points

Starting from a uniform prior, what is the updated belief if the spacecraft takes action a_{CA} and observes a positive spectrometer reading? Round all values in your answer to two decimal places and maintain the following order in your belief vector: $s_{HH}, s_{HL}, s_{LH}, s_{LL}, s_{EM}$.

quiz3_2.png Download



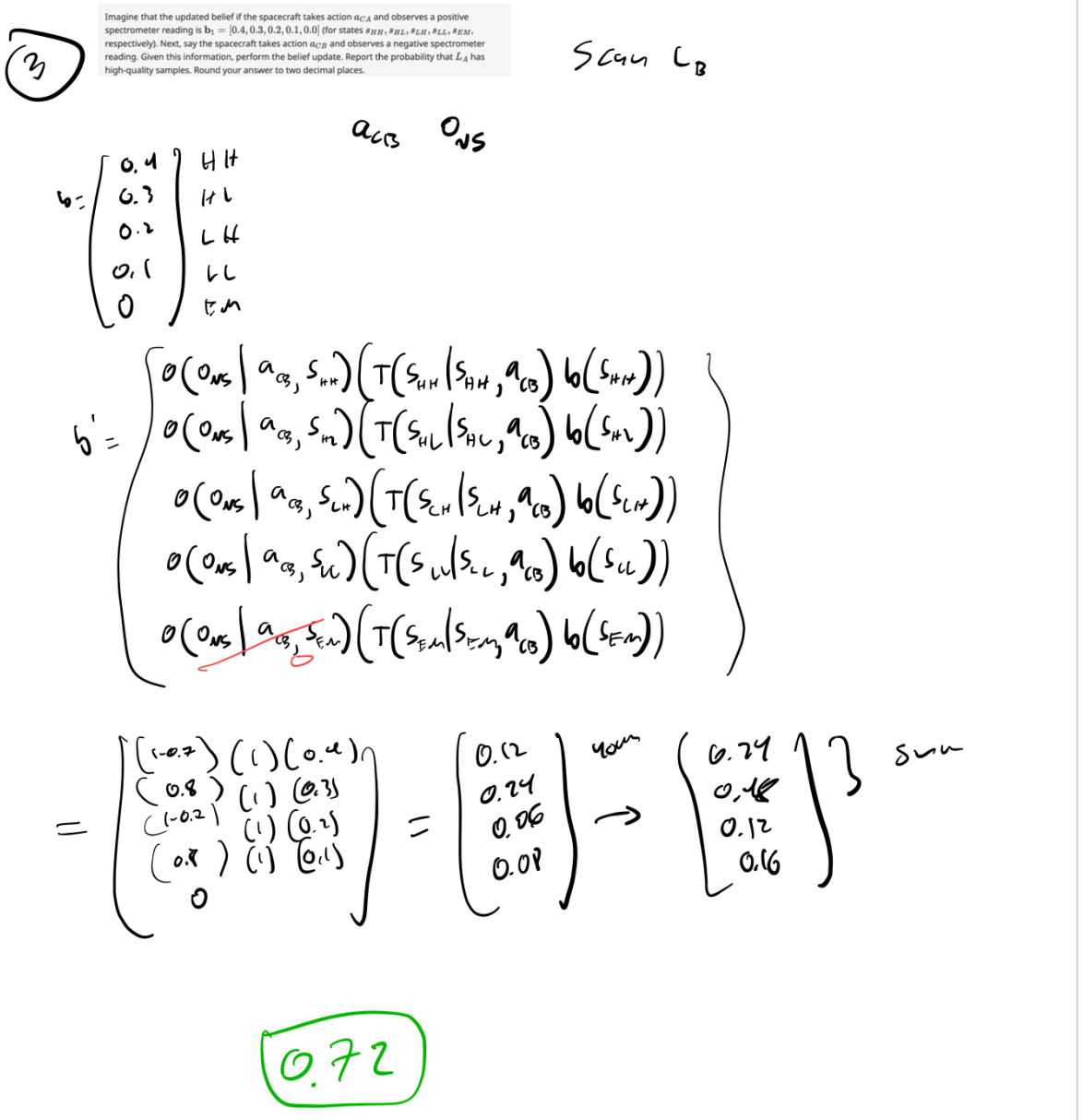
Q3

5 Points

Imagine that the updated belief if the spacecraft takes action a_{CA} and observes a positive spectrometer reading is $\mathbf{b}_1 = [0.4, 0.3, 0.2, 0.1, 0.0]$ (for states $s_{HH}, s_{HL}, s_{LH}, s_{LL}, s_{EM}$, respectively). Next, say the spacecraft takes action a_{CB} and observes a negative spectrometer reading. Given this information, perform the belief update. Report the probability that L_A has high-quality samples. Round your answer to two decimal places.

quiz3_3.png

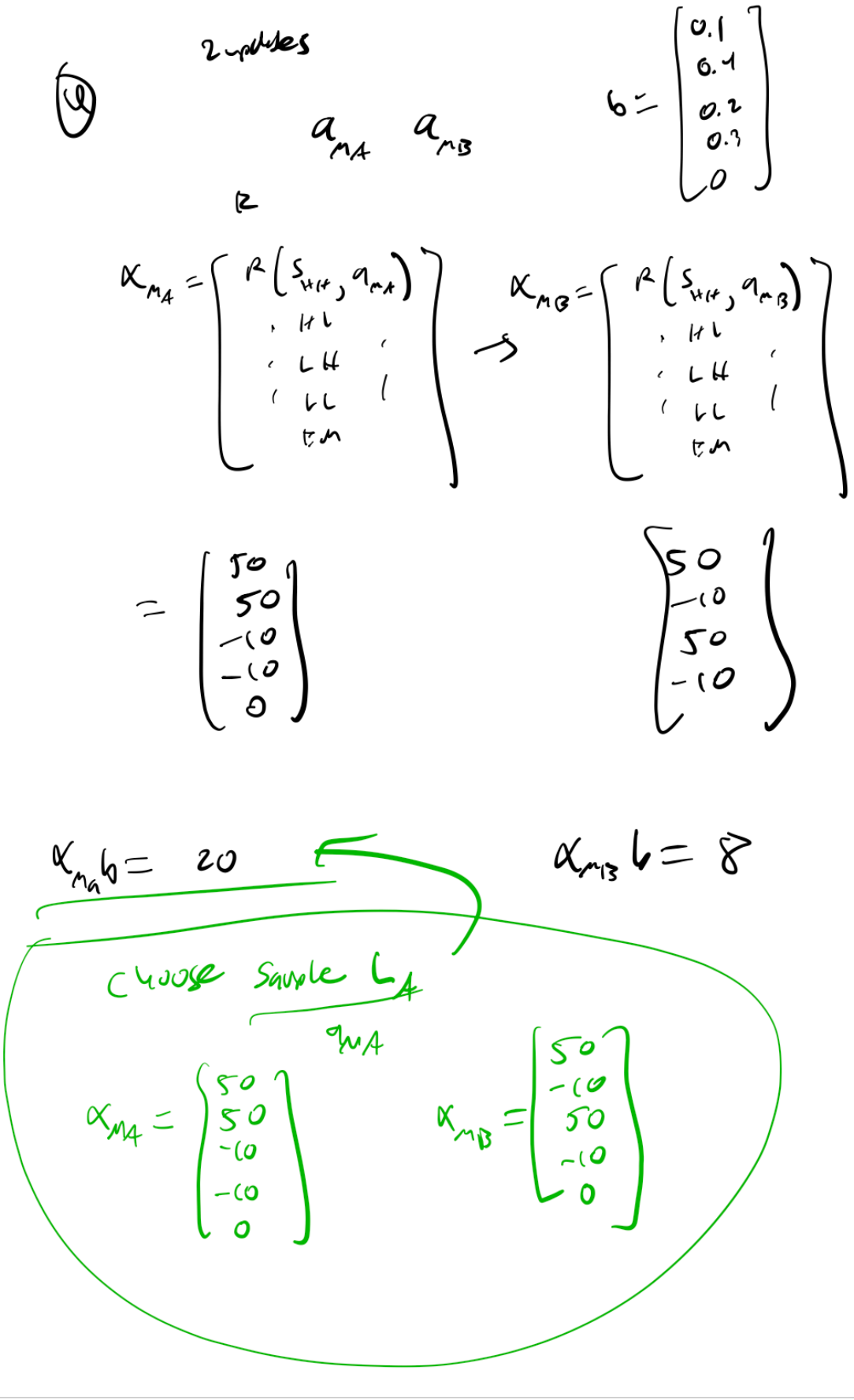
Download



Q4
5 Points

After the two belief updates above, imagine you arrive at belief vector $\mathbf{b}_2 = [0.1, 0.4, 0.2, 0.3, 0.0]$ (for states $s_{HH}, s_{HL}, s_{LH}, s_{LL}, s_{EM}$, respectively). With this belief, calculate the rewards associated with sampling L_A and sampling L_B. Provide: (1) each expected reward and (2) a recommendation for which site to sample.

quiz3_4.png Download



Q5

5 Points

For this question only, suppose we recalibrate our spectrometer so that the probability of the spectrometer giving a positive reading for a site with low-quality samples (a false positive) is only 0.1. If the spacecraft takes action a_{CA} and receives a positive reading from the recalibrated spectrometer, what is the probability that L_A has high-quality samples? Assume a uniform prior. Round your answer to two decimal places.

quiz3_5.png Download

5

a_{CA} O_{PS}

given L_A

$$b = \begin{bmatrix} 0.2 \\ 0.2 \\ 0.1 \\ 0.2 \\ 0.2 \end{bmatrix}$$

$$b' = \begin{pmatrix} O(O_{PS} \mid a_{CA}, S_{HH})\left(T(S_{HH} \mid S_{HH}, a_{CA})\, b(S_{HH})\right) \\ O(O_{PS} \mid a_{CA}, S_{HL})\left(T(S_{HL} \mid S_{HL}, a_{CA})\, b(S_{HL})\right) \\ O(O_{PS} \mid a_{CA}, S_{LH})\left(T(S_{LH} \mid S_{LH}, a_{CA})\, b(S_{LH})\right) \\ O(O_{PS} \mid a_{CA}, S_{LL})\left(T(S_{LL} \mid S_{LL}, a_{CA})\, b(S_{LL})\right) \\ O(O_{PS} \mid a_{CA}, S_{EM})\left(T(S_{EM} \mid S_{EM}, a_{CA})\, b(S_{EM})\right) \end{pmatrix}$$

$O(O_i \mid S_i) = 0.2$ (i)

PDS given low $O(O_i \mid S_i) = 0.1$

$$= \begin{pmatrix} (0.7)(1)(0.2) \\ (0.7)(1)(0.2) \\ (0.1)(1)(0.2) \\ (0.1)(1)(0.2) \\ (0)(0.2) \end{pmatrix} = \begin{bmatrix} 0.14 \\ 0.14 \\ 0.02 \\ 0.02 \\ 0 \end{bmatrix} \rightarrow b' = \begin{bmatrix} 0.4375 \\ 0.4375 \\ 0.0625 \\ 0.0625 \\ 0 \end{bmatrix}$$

- Observations:
  - If we scan (choose a_{CA} or a_{CB}), we observe either a positive or negative reading from the spectrometer.
  - If we sample (choose a_{MA} or a_{MB}), we observe either a biosignature detection or no detection.
  - If we have ended the mission (a_{EM}), we receive no observation information at all.
  - The probability of a positive reading for a site with high-quality samples is 0.7.

1

0.88

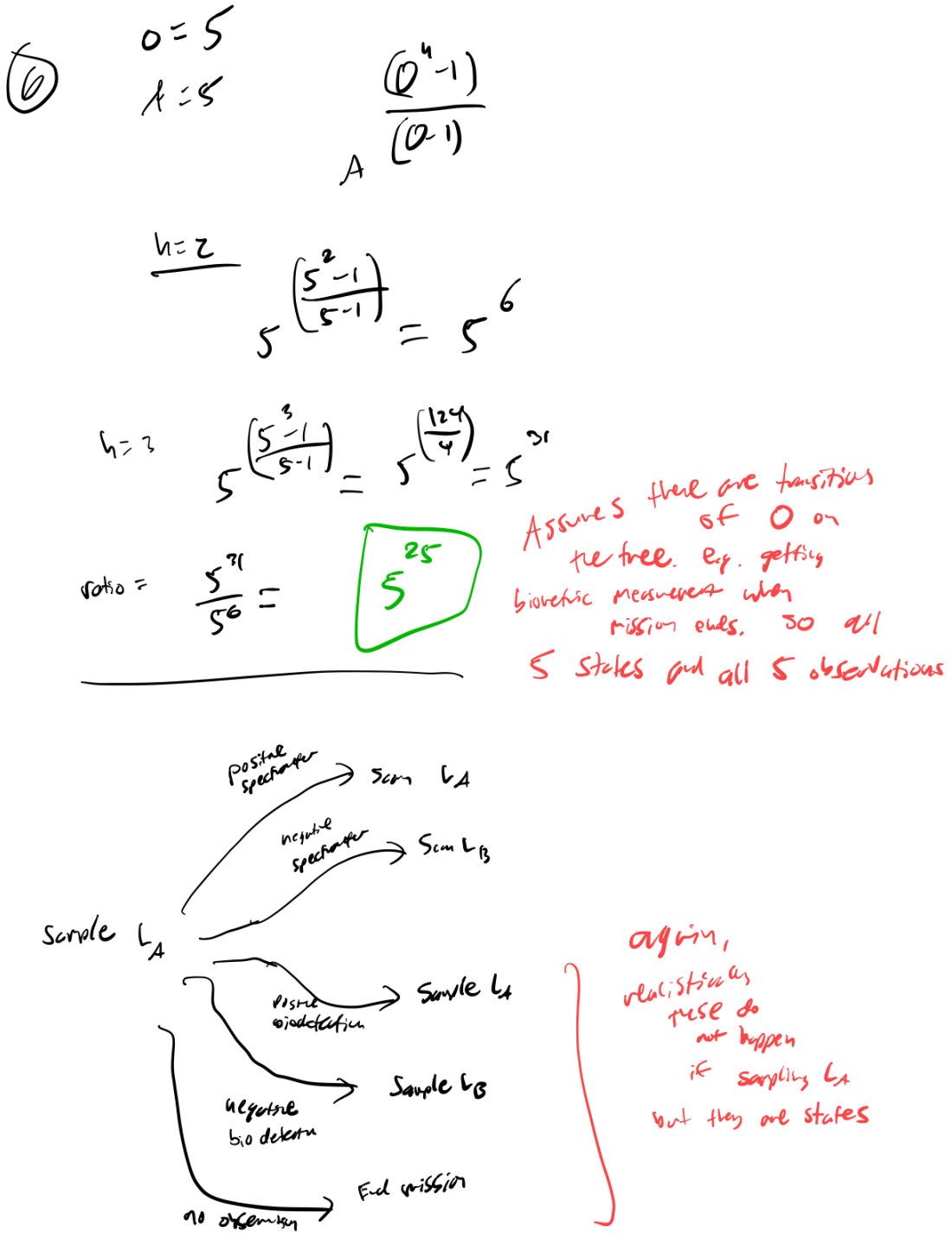
- The probability of a negative reading for a site with low-quality samples is 0.8. 0.9
- The probability of no biosignature detection in a high-quality sample is 0.05.
- The probability of a biosignature detection in a low-quality sample is 0.05.

Q6

5 Points

What is the ratio of the number of possible 3-step plans to the number of possible 2-step plans? In addition to your numerical answer, draw one of the possible 2-step conditional plans, labeling each node with an action and each edge with an observation.

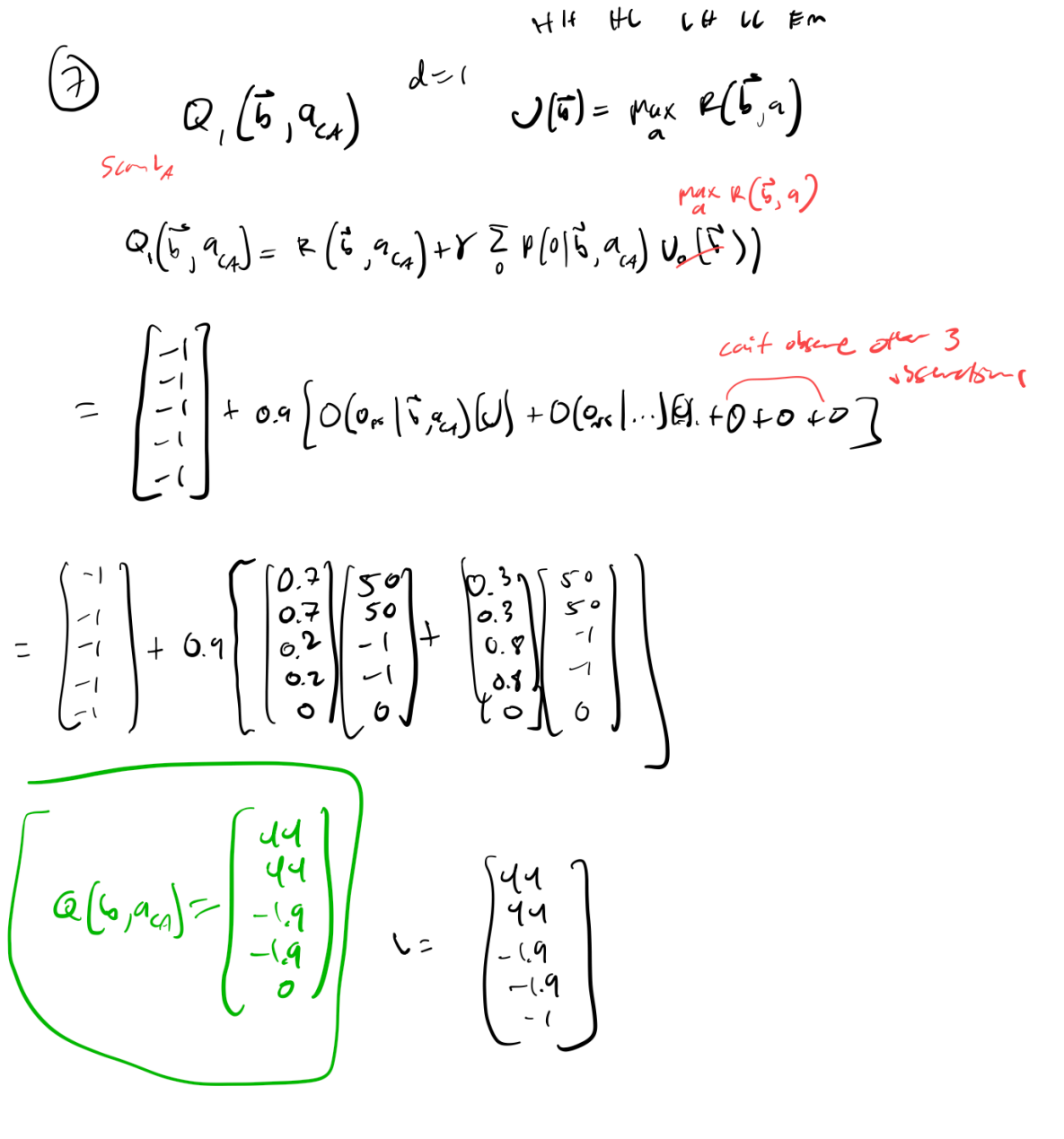
quiz3_6.png Download



Q7
5 Points

Assuming a uniform prior, compute $Q_1(\mathbf{b}, a_{CA})$ using forward search to depth 1. Let $U(\mathbf{b}) = \max_a R(\mathbf{b}, a)$. Round your answer to two decimal places and assume the following order for the belief vector: $s_{HH}, s_{HL}, s_{LH}, s_{LL}, s_{EM}$.

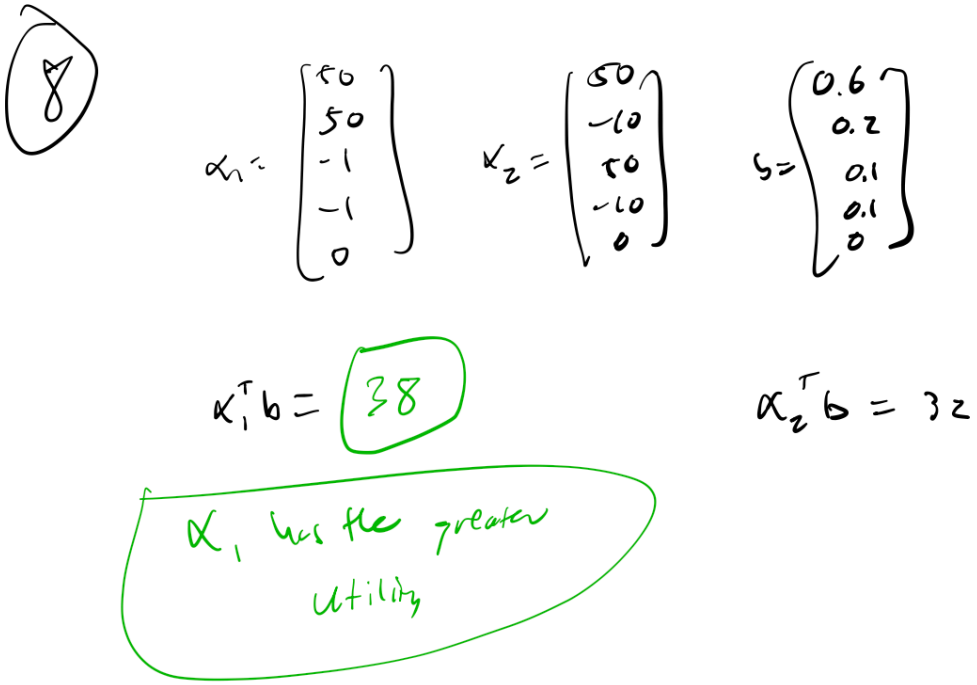
quiz3_7.png Download



Q8
5 Points

Suppose we have two alpha vectors, $\alpha_1 = [50, 50, -10, -10, 0]$ and $\alpha_2 = [50, -10, 50, -10, 0]$. Assume states are ordered $s_{HH}, s_{HL}, s_{LH}, s_{LL}$ and s_{EM}. From belief $\mathbf{b} = [0.6, 0.2, 0.1, 0.1, 0.0]$, calculate the expected value of each alpha vector. Provide a one sentence interpretation of your results that identifies which policy you would choose and why.

quiz3_8.png Download



Q9
5 Points

In the context of our robotic sampling problem, describe why QMDP can overestimate the optimal value function and how the fast informed bound provides a tighter upper bound. Keep your answer within 2-3 sentences and be sure to include any relevant context from the problem formulation.

A QMDP will know whether a sample is high quality, so it will always produce the optimal action for a reward of 50. Since a POMDP holds a vector of beliefs, fast informed bound propagates its alpha vector with a weighted average by the probability of actually being in that state, so it may not pick a reward of 50 if it believes there is a small chance that the rover is actually in a high quality state.

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Q10
5 Points

You have taken on the role of mission planner to inform the search for biosignatures on POMDPlanet. You have several days to compute the optimal policy for a 20-step mission with 5 potential sampling locations. In 2-3 sentences, describe a benefit of using point-based value iteration for this task over QMDP. Again, be sure to include any relevant context from the problem formulation.

For 20 steps with 5 sampling locations, which is much more states and steps, the spacecraft can take the time to run many alpha vectors and belief states (per point-based value iteration) than just 1 per action (like in QMDP).

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Q11
5 Points

Suppose we are performing MCTS with the UCB1 exploration heuristic. Our exploration factor is $c = 1$ and we are at the root node with a uniform prior. After multiple simulations, we have the following:

Action	N(h,a)	Q(h,a)
a_{CA}	4	21.0
a_{CB}	3	18.5
a_{MA}	2	24.0
a_{MB}	1	19.0
a_{EM}	1	0.0

Given our choice of heuristic, calculate the scores for each action and report which action the spacecraft should take and its corresponding score. Use natural log ($\ln$) when applicable and round your answer to two decimal places.

quiz3_11.png

Download

11



Q11
5 Points

Suppose we are performing MCTS with the UCB1 exploration heuristic. Our exploration factor is $c = 1$ and we are at the root node with a uniform prior. After multiple simulations, we have the following:

Action	N(h,a)	Q(h,a)
a_{CA}	4	21.0
a_{CB}	3	18.5
a_{MA}	2	24.0
a_{MB}	1	19.0
a_{EM}	1	0.0

Given our choice of heuristic, calculate the scores for each action and report which action the spacecraft should take and its corresponding score. Use natural log (ln) when applicable and round your answer to two decimal places.

$$Q(h,a) + c\sqrt{\frac{\log N(h)}{N(h,a)}}$$

α_{CA}: $21 + (1)\sqrt{\frac{\log(11)}{4}} = 21.77$

α_{CB}: $18.5 + (1)\sqrt{\frac{\log(11)}{3}} = 19.39$

α_{MA}: $24 + (1)\sqrt{\frac{\log(11)}{2}} = 25.10$

α_{MB}: $19 + (1)\sqrt{\frac{\log(11)}{1}} = 20.55$

α_{EM}: $0 + (1)\sqrt{\frac{\log(11)}{1}} = 1.55$

a_{MA}